

MSE 311 Thermodynamics of materials-Homework 8

Name:

Student ID:

Please hand in this homework before class on November 10th.

1. The vapor pressure of solid NaF varies with temperature as

$$\ln p(\text{atm}) = \frac{-34,450}{T} - 2.01 \ln T + 33.74$$

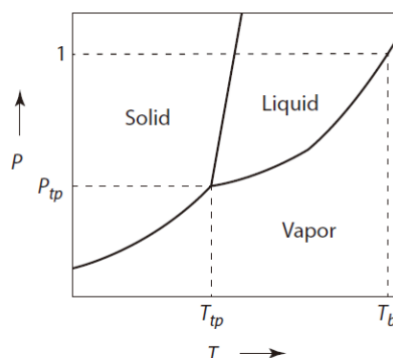
and the vapor pressure of liquid NaF varies with temperature as

$$\ln p(\text{atm}) = \frac{-31,090}{T} - 2.52 \ln T + 34.66$$

Calculate

- The normal boiling temperature of NaF;
- The temperature and pressure at the triple point;
- The molar enthalpy of evaporation of NaF at its normal boiling temperature;
- The molar enthalpy of melting of NaF at the triple point;
- The difference between the constant-pressure molar heat capacities of liquid and solid NaF.

The phase diagram is shown schematically as following.



- The normal boiling temperature, T_b , is defined as that temperature at which the saturated vapor pressure of the liquid is 1 atm. Thus, from the equation for the vapor pressure of the liquid, T_b is

$$\ln(1) = 0 = -\frac{31,090}{T_b} - 2.52 \ln T_b + 34.66$$

which has the solution

$$T_b = 2006 \text{ K}$$

2. The saturated vapor pressures for the solid and liquid phases intersect at the triple point. Thus, at the temperature, T_{tp} , of the triple point

$$-\frac{34,450}{T_{tp}} - 2.01 \ln T_{tp} + 33.74 = -\frac{31,090}{T_{tp}} - 2.52 \ln T_{tp} + 34.66$$

which has the solution

$$T_p = 1239 \text{ K}$$

The triple-point pressure is then calculated from the equation for the vapor pressure of the solid as

$$p = \exp\left(-\frac{34,450}{1239} - 2.01 \ln 1239 + 33.74\right) = 2.29 \times 10^{-4} \text{ atm}$$

or from the equation for the vapor pressure for the liquid as

$$p = \exp\left(-\frac{31,090}{1239} - 2.52 \ln 1239 + 34.66\right) = 2.29 \times 10^{-4} \text{ atm}$$

3. For vapor in equilibrium with the liquid:

$$\begin{aligned} \ln p \text{ (atm)} &= -\frac{31,090}{T} - 2.52 \ln T + 34.66 \\ \frac{d \ln p}{dT} &= \frac{\Delta H}{RT^2} = \frac{31,090}{T^2} - \frac{2.52}{T} \end{aligned}$$

Thus,

$$\Delta H_{(l \rightarrow v)} = (31,090 \times 8.3144) - (2.52 \times 8.3144)T = 258,500 - 20.95T$$

and, at the normal boiling temperature of 2006 K,

$$\Delta H_{(l \rightarrow v)} = 258,500 - 20.95 \times 2006 = 216,500 \text{ J}$$

4. For vapor in equilibrium with the solid:

$$\ln p \text{ (atm)} = -\frac{34,450}{T} - 2.01 \ln T + 33.74$$

Thus,

$$\begin{aligned} \Delta H_{(s \rightarrow v)} &= (34,450 \times 8.3144) - (2.01 \times 8.3144)T \\ &= 286,400 - 16.71T \text{ J} \end{aligned}$$

Near the triple point,

$$\Delta H_{(s \rightarrow l)} + \Delta H_{(l \rightarrow v)} = \Delta H_{(s \rightarrow v)}$$

and thus,

$$\begin{aligned} \Delta H_{(s \rightarrow l)} &= 286,400 - 16.71T - 258,500 + 20.95T \\ &= 27,900 + 4.24T \end{aligned}$$

At the triple point,

$$\Delta H_{(s \rightarrow l)} = 27,900 + (4.24 \times 1239) = 33,150 \text{ J}$$

$$5 \quad \Delta H_{(s \rightarrow l)} = 27,900 + 4.24T$$

$$\begin{aligned} \frac{d\Delta H}{dT} &= \Delta c_p = 4.24 \text{ J/K} \\ &= c_p^L - c_p^S > 0 \end{aligned}$$

2. Carbon has the following three allotropes: graphite, diamond, and a metallic form called solid III. Graphite is the stable form of 1000 K and 1 atm pressure, and increasing the pressure on graphite at temperatures less than 1440 K causes the transformation of graphite to diamond and then the transformation of diamond to solid III. Calculate the pressure which, when applied to one mole graphite at 1000 K, causes the transformation of graphite to diamond, given

$$H_{1000K, \text{graphite}} - H_{1000K, \text{diamond}} = -1896J$$

$$S_{1000K, \text{graphite}} = 5.74J / K$$

$$S_{1000K, \text{diamond}} = 2.36J / K$$

Density of graphite at 1000K: 2.62 g/cm^3

Density of diamond at 1000K: 3.52 g/cm^3

For the transformation graphite $\rightarrow$ diamond at 1000 K:

$$\Delta G = \Delta H - T\Delta S = 1896 - 1000 \times (2.36 - 5.74) = 5276J$$

For the transformation of graphite to diamond at any temperature T:

$$\Delta V = V_{\text{diamond}} - V_{\text{graphite}} = \frac{12}{3.52} - \frac{12}{2.62} = -1.17 \text{ cm}^3 / \text{mole}$$

Equilibrium between graphite and diamond at 298 K requires that $\Delta G_{\text{graphite} \rightarrow \text{diamond}}$ be zero. Since

$$\Delta G(P, T = 1000) = \Delta G(P = 1, T = 1000) + \int_1^P \Delta V dP$$

If the difference between the isothermal compressibilities of the two phases is negligibly small (i.e., if the influence of pressure on ΔV can be ignored), then

$$-5276 = -1.17 \times 0.1013(P - 1)$$

$$P = \frac{5276}{1.17 \times 0.1013} + 1 = 44516 \text{ atm}$$

3. Please look for a material that has the following properties expect for water.

